

EN. 705.743 ChatGPT from Scratch: Building and Training Large Language Models

Final Project Rubric

Category	Full Credit	Strong	Partial	Minimal	Max Pts
TECHNICAL CORRECTNESS Implementation & Reproducibility	- Core implementation is fully correct and end-to-end functional - All results are reproducible from submitted code and instructions - All components work as intended with no significant bugs	- Implementation is mostly correct with minor bugs - Bugs do not affect overall conclusions or reproducibility - All major components are present and functional	- Core implementation works but has notable errors or missing components - Results may not be fully reproducible without extra effort	- Implementation is incomplete or incorrect - Code does not run or results cannot be reproduced	15
CONCEPTUAL UNDERSTANDING Design Decisions & Course Concepts	- Design decisions are clearly motivated and grounded in course concepts - Write-up and/or Q&A responses are confident and accurate - Student clearly explains *why* choices were made, not just what was done	- Understanding is evident with minor gaps in justification - Most design decisions are explained; terminology is mostly accurate	- Understanding is superficial; choices are stated but not explained - Little connection made between design choices and course concepts	- Little or no evidence of conceptual understanding - Relies on copied code or templates without explanation	20
EXPERIMENTS & ANALYSIS Rigor & Interpretation of Results	- Multiple well-designed experiments with clear analysis - Results are interpreted in context; failure modes are discussed - Ablations or comparisons are included and meaningfully interpreted	- Experiments are present and results are clearly reported - Analysis is reasonable but could go deeper or cover more cases	- Limited experimentation; results reported but not interpreted - No ablations or comparisons; conclusions are unsupported	- No meaningful experiments conducted - Results are missing or conclusions are entirely unsupported	10

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CODE QUALITY Readability, Structure & Documentation	- Code is clean, modular, and thoroughly documented - A peer could install dependencies and reproduce results with no guidance - README or equivalent is clear and complete	- Code is generally readable with adequate inline comments - Minor clarity issues; a peer could reproduce with minimal effort	- Code runs but is difficult to follow; limited documentation - Reproduction requires significant effort or instructor help	Code is disorganized, undocumented, or cannot be run by the grader	15
WRITE-UP / REPORT Communication & Documentation	- Report is polished and complete: problem statement, related work, methodology, results, discussion, and limitations - Includes a block diagram or architecture figure where the model design warrants it; figures are clearly labeled and referenced	- Report covers all main sections with minor gaps in depth or clarity - Architecture diagram is present but could be clearer or better integrated - Figures are present but may lack labels or in-text references	- Report is incomplete or unclear in one or more key sections - Architecture diagram is missing where it would clearly aid understanding - Results are presented without sufficient context or discussion	- Report is missing major sections or is too brief to assess the work - No figures or diagrams provided; conclusions are absent	20
PRESENTATION Clarity, Timing & Q&A	- Well-organized, within the time limit, and visuals effectively support the narrative - Student handles audience questions confidently and accurately	- Mostly clear with minor pacing or clarity issues - Questions answered adequately with only minor gaps	- Presentation is hard to follow or significantly over/under time - Student struggles to answer questions about their own work	Presentation is missing, incoherent, or student cannot answer basic questions	20
Total					100